# Developer Codebase & Architecture Guide

Comprehensive Technical Reference: File-by-File Functional Breakdown, Responsibilities, & System Features

**DEVELOPER PERSPECTIVE OVERVIEW:**
ChairCare is an enterprise on-demand ergonomic chair repair and calibration platform built with a unified full-stack architecture: a cross-platform **React Native / Expo** client supporting iOS, Android, and Web, backed by a **Node.js TypeScript Express** REST API and a **PostgreSQL** relational database. The system coordinates three authenticated actor roles: **Customers** (booking, telemetry, live Razorpay payments), **Field Technicians** (queue dispatch, turn-by-turn navigation, PIN verification, offline photo logs, payout ledger), and **SuperAdmins** (live operations command, dispute escrow arbitration, KYC approvals, dynamic rate cards). This document provides an exhaustive, file-by-file audit of every key module, component, controller, and service in the codebase.

| | |
|---|---|
| **Frontend Framework: React Native 0.76 / Expo SDK 52 (TypeScript)** | **Backend Stack: Node.js 20, Express, TypeScript** |
| **Database: PostgreSQL (Pooled via pg)** | **Payment Gateway: Razorpay API (Configured via .env)** |
| **Single Payment Rule: Strict 1-Order-1-Payment constraint enforced** | **Security Signoff: 4-Digit Customer PIN Verification on completion** |
| **Dynamic Platform Fee: 10% dynamic platform fee configured by Admin & itemized on booking** | **Direct UPI Payments: Technician scannable UPI QR code generation and instant customer payments** |

## SECTION 1: ROOT CONFIGURATIONS & TOOLING

**8 Core Files**

*Top-level application entry points, mobile build manifestos, bundle definitions, and project configuration.*

### `FILE: App.tsx`

**PRIMARY ROLE:** Main root entry point of the React Native / Expo application.

**KEY FEATURES & CODE LOGIC:**
- Bootstraps safe area context and theme provider hierarchy.
- Wraps application in AuthProvider, BookingProvider, TechnicianProvider, NetworkProvider, and IntroProvider.
- Integrates NetworkStatusBoundary for global offline/online connectivity banner.
- Loads RootNavigator for role-based conditional navigation routing.

**DEVELOPER NOTE:** Entry point for all mobile app bundles (iOS, Android, and Web). Ensure new global providers are mounted in the Provider tree here.

### `FILE: index.js`

**PRIMARY ROLE:** Expo / React Native application register script.

**KEY FEATURES & CODE LOGIC:**
- Registers the root React component with AppRegistry.
- Ensures seamless entry across native iOS, Android Hermes, and Web runtimes.

**DEVELOPER NOTE:** Invoked automatically by the Expo bundler (`expo start`).

### FILE: app.json

**PRIMARY ROLE:** Expo build & application manifest configuration.

**KEY FEATURES & CODE LOGIC:**
• Defines application slug, display name ('ChairCare'), orientation, and icons.
• Configures splash screen settings, Android permissions, and iOS bundle identifier.
• Declares Expo plugins for camera, location, and storage permissions.

**DEVELOPER NOTE:** Target for Expo Application Services (EAS) builds and OTA updates.

### FILE: eas.json

**PRIMARY ROLE:** Expo Application Services build profiles.

**KEY FEATURES & CODE LOGIC:**
• Defines development, preview, and production APK/IPA build targets.
• Configures environment variable injection and build credentials.

**DEVELOPER NOTE:** Used when running `eas build --profile production` for store submissions.

### FILE: package.json

**PRIMARY ROLE:** Client project dependencies and script definitions.

**KEY FEATURES & CODE LOGIC:**
• Declares React 18, React Native 0.76, Expo SDK 52 dependencies.
• Configures navigation (@react-navigation/native, native-stack, bottom-tabs).
• Includes Lucide icons, Expo Vector Icons, Linear Gradient, and image processing.

**DEVELOPER NOTE:** Run `npm install` inside root directory for mobile/web dependencies.

### FILE: tsconfig.json

**PRIMARY ROLE:** TypeScript compiler settings for the frontend application.

**KEY FEATURES & CODE LOGIC:**
• Extends `expo/tsconfig.base` for strict typed React Native code.
• Configures JSX transform, path aliases, and strict type validation.

**DEVELOPER NOTE:** Validated via `npx tsc --noEmit` before any release commit.

### FILE: .env & .env.example

**PRIMARY ROLE:** Environment variables and gateway credential storage.

**KEY FEATURES & CODE LOGIC:**
• Configures `EXPO_PUBLIC_API_URL` (e.g. https://api.yourdomain.com/api/v1).
• Stores `EXPO_PUBLIC_RAZORPAY_KEY_ID` (Razorpay Public Key ID).
• Sets active runtime mode (production vs test).

**DEVELOPER NOTE:** Secrets should never be committed to git. Use EAS secrets for remote builds.

### FILE: DEPLOYMENT.md

**PRIMARY ROLE:** Production deployment operational runbook.

**KEY FEATURES & CODE LOGIC:**
• Full steps for deploying backend to Render.com and frontend to Expo / Vercel.
• PostgreSQL schema migration scripts and seed instructions.
• SSL, CORS, domain mapping, and Cloudflare tunnel instructions.

**DEVELOPER NOTE:** Read this before deploying updates to live production environments.

## SECTION 2: NAVIGATION ARCHITECTURE (SRC/NAVIGATION/) 5 Core Files

*Navigation stacks and tab bars routing users, technicians, and superadmins across the application.*

### FILE: src/navigation/RootNavigator.tsx

**PRIMARY ROLE:** Master top-level routing container switching between application flows.

**KEY FEATURES & CODE LOGIC:**
• Monitors active authenticated user role (`customer`, `technician`, `admin`, `guest`).
• Renders IntroNavigator for first-time onboarding visitors.
• Mounts CustomerNavigator, TechnicianNavigator, or AdminNavigator based on credentials.
• Provides deep linking configuration and DevRoleSwitcher integration.

**DEVELOPER NOTE:** Changes here alter how users are directed after authentication or app restart.

### FILE: src/navigation/CustomerNavigator.tsx

**PRIMARY ROLE:** Primary multi-tab and stack navigator for end-customers.

**KEY FEATURES & CODE LOGIC:**
• Bottom Tab Bar: Services Available, Active Service, My Requests, Notifications, Profile.
• Stack Screens: 5-step interactive booking flow, Live Map Tracking, Payment Invoice, Payment Success, Review Modal, and Chat.
• Dynamic badge indicators on requests and notifications tabs.

**DEVELOPER NOTE:** All customer screen route parameters are strongly typed in `types/navigation.ts`.

### FILE: src/navigation/TechnicianNavigator.tsx

**PRIMARY ROLE:** Certified Field Specialist operations navigator.

**KEY FEATURES & CODE LOGIC:**
• Bottom Tab Bar: Job Requests Queue, Active Job Execution, Earnings Ledger, History, Profile.
• Stack Screens: Job Request Details, Navigation Route Map, PIN Verification, Additional Work Approval, Completion Sign-off.
• Guarded routing based on technician KYC verification status.

**DEVELOPER NOTE:** Technicians who are unverified or suspended are redirected to account status screens.

### FILE: src/navigation/AdminNavigator.tsx

**PRIMARY ROLE:** Operations & SuperAdmin command center navigator.

**KEY FEATURES & CODE LOGIC:**
• Bottom Tab Bar: Live Operations, Service Dispatch, Technicians Fleet, Payments & Refunds, Platform Settings.
• Stack Screens: Dispute Arbitration, Review Moderation, Audit Log Export, Customer Management, Manual Technician Assignment.
• Secured with high-privilege admin role checks.

**DEVELOPER NOTE:** Allows real-time management of all orders, payouts, and system configurations.

### FILE: src/navigation/IntroNavigator.tsx

**PRIMARY ROLE:** First-time user onboarding carousel stack.

**KEY FEATURES & CODE LOGIC:**
• Smooth walkthrough: Instant Ergonomic Repair, Certified Specialists, Live Telemetry, BIFMA Quality Standards.
• Persists `hasSeenIntro` flag in AsyncStorage to bypass on subsequent app visits.

**DEVELOPER NOTE:** Users who click 'Get Started' are transitioned directly into the customer portal.

## SECTION 3: GLOBAL STATE MANAGEMENT (SRC/CONTEXT/) 6 Core Files

*React Context providers managing authentication, real-time booking draft, technician status, and theme state.*

### FILE: src/context/AuthContext.tsx

**PRIMARY ROLE:** Authentication, JWT tokens, session persistence, and user profile state.

**KEY FEATURES & CODE LOGIC:**
• Manages `user`, `token`, `role`, and `isAuthenticated` state.
• Persists authentication tokens securely in AsyncStorage.
• Provides `login()`, `register()`, `logout()`, `updateUser()`, and dev role switching.
• Auto-syncs user details with backend user profile on startup.

**DEVELOPER NOTE:** All API calls in `apiClient.ts` read the bearer token from this session storage.

### FILE: src/context/BookingContext.tsx

**PRIMARY ROLE:** Interactive multi-step service booking draft state and pricing engine.

**KEY FEATURES & CODE LOGIC:**
• Maintains draft state across 5 booking steps (chair model, problem, address, date/time, price).
• Dynamic Platform Fee: Applies 10% platform fee calculation to base service pricing with itemized customer breakdown.
• Calculates service charges, spare parts estimates, and final grand total.
• Provides `updateDraft()`, `resetDraft()`, and final order placement handlers.
• Caches drafted details locally so user doesn't lose inputs if interrupted.

**DEVELOPER NOTE:** Passes accumulated draft details and calculated fees to `customerApi.createServiceRequest()`.

### FILE: src/context/TechnicianContext.tsx

**PRIMARY ROLE:** Active technician telemetry, dynamic UPI payment settings, and job state management.

**KEY FEATURES & CODE LOGIC:**
• Manages active job status, accepted request details, and live coordinates.
• Direct UPI & QR Settings: Manages dynamic `upiId`, `upiPhone`, and `updateUpiSettings()` synced with backend.
• Direct Customer QR Collections: Maintains `qrCollections` array and `recordQrPayment()` with persistent AsyncStorage caching.
• Provides `toggleAvailability()` for duty on/off toggle.
• Persists photo diagnostic logs and service timer during active repairs.

**DEVELOPER NOTE:** Active repair state, QR transaction logs, and UPI credentials persist across sessions.

### FILE: src/context/ThemeContext.tsx

**PRIMARY ROLE:** Application theme provider and dark/light mode toggle.

**KEY FEATURES & CODE LOGIC:**
• Provides active color palette, typography tokens, and spacing metrics.
• Allows on-the-fly toggling between sleek dark mode and high-contrast light mode.
• Persists user theme preferences.

**DEVELOPER NOTE:** Consumes design tokens declared in `src/constants/theme.ts`.

### FILE: src/context/NetworkContext.tsx

**PRIMARY ROLE:** Real-time network connection monitoring.

**KEY FEATURES & CODE LOGIC:**
• Polls network status and listens to browser/native connection state.
• Exposes `isConnected` and `isInternetReachable` booleans to UI components.
• Triggers offline fallback alerts when server is unreachable.

**DEVELOPER NOTE:** Connected to `NetworkStatusBoundary.tsx` to display top warning banner.

### FILE: src/context/IntroContext.tsx

**PRIMARY ROLE:** First-time visitor intro status management.

**KEY FEATURES & CODE LOGIC:**
• Tracks whether user has completed the 4-slide introductory tutorial.
• Provides `completeIntro()` and `resetIntro()` methods.

**DEVELOPER NOTE:** Integrated with `utils/introStorage.ts` for AsyncStorage persistence.

## SECTION 4: CLIENT SERVICES & API GATEWAY (SRC/SERVICES/) 10 Core Files

*Network API clients, Razorpay payment triggers, geocoding, voice calling, and device utilities.*

### FILE: src/services/api/apiClient.ts

**PRIMARY ROLE:** Central Axios HTTP client configured with bearer token interceptors and base URL.

**KEY FEATURES & CODE LOGIC:**
• Automatically attaches JWT `Authorization: Bearer ` to all outbound requests.
• Standardizes error handling with `sanitizeUserErrorMessage()`.
• Provides clean response type contracts (`ApiResponse`).
• Configured with 15-second timeouts and CORS headers.

**DEVELOPER NOTE:** All domain-specific API files route their HTTP calls through this shared client.

### FILE: src/services/api/paymentApi.ts

**PRIMARY ROLE:** Payment gateway client for order creation, config retrieval, and signature verification.

**KEY FEATURES & CODE LOGIC:**
• Calls `GET /api/v1/payments/config` to get active Razorpay key ID & environment.
• Calls `POST /api/v1/payments/create-order` to generate a cryptographic Razorpay Order ID.
• Calls `POST /api/v1/payments/verify` with HMAC-SHA256 signature and payment ID.
• Contains single-payment checks returning `alreadyPaid: true` if already settled.

**DEVELOPER NOTE:** Integrates directly with `RazorpayCheckoutModal.tsx` and `PaymentInvoiceChaircareMobileScreen.tsx`.

### FILE: src/services/api/customerApi.ts

**PRIMARY ROLE:** Customer lifecycle API methods.

**KEY FEATURES & CODE LOGIC:**
• Creates service requests with UUID sanitization (`createServiceRequest`).
• Fetches active requests (`getMyRequests`), request details (`getRequestDetails`).
• Cancels requests, submits customer reviews, and registers dispute tickets.

**DEVELOPER NOTE:** Sanitizes non-UUID service IDs into valid service codes for backend compatibility.

### FILE: src/services/api/technicianApi.ts

**PRIMARY ROLE:** Technician job dispatch and execution API client.

**KEY FEATURES & CODE LOGIC:**
• Fetches nearby unassigned queue jobs (`getAvailableJobs`).
• Accepts requests, updates job status (`arrived`, `in_progress`, `completed`).
• Verifies customer 4-digit Security PIN (`verifySecurityPin`).
• Uploads pre/post repair photo logs and requests additional work approval.

**DEVELOPER NOTE:** PIN verification is mandatory before a technician can transition an order to completed.

### FILE: src/services/api/adminApi.ts

**PRIMARY ROLE:** Operations & SuperAdmin management API client.

**KEY FEATURES & CODE LOGIC:**
• Queries live platform metrics, operational queue, and revenue totals.
• Approves/suspends technician KYC and documents.
• Issues customer refunds and resolves customer dispute arbitration.
• Exports CSV audit logs and modifies dynamic service rate cards.

**DEVELOPER NOTE:** Restricted to users with role `'admin'`.

### FILE: src/services/api/authApi.ts

**PRIMARY ROLE:** User authentication and profile management API.

**KEY FEATURES & CODE LOGIC:**
• Handles customer & technician login and registration.
• Refreshes user profile, updates passwords, and manages push notification tokens.

**DEVELOPER NOTE:** Returns JWT token with user object upon successful authentication.

### FILE: src/services/api/chatApi.ts

**PRIMARY ROLE:** Real-time in-app messaging between customer, technician, and support.

**KEY FEATURES & CODE LOGIC:**
• Fetches conversation message history by request ID or support ticket ID.
• Sends text messages and base64-encoded attachment images.
• Marks messages as read.

**DEVELOPER NOTE:** Used by `ChatWithTechnicianChaircareMobileScreen` and `SupportConversationChaircareMobileScreen`.

### FILE: src/services/phone/phoneCallService.ts

**PRIMARY ROLE:** Voice calling and communication bridge utility.

**KEY FEATURES & CODE LOGIC:**
• Triggers native phone dialer via `tel:` URL scheme.
• Opens interactive in-app simulated voice call modal on web.
• Supports customer-to-technician direct calling.

**DEVELOPER NOTE:** Ensures telephony functionality works smoothly across native iOS/Android and web.

### FILE: src/services/location/geocodingService.ts

**PRIMARY ROLE:** Geocoding and distance calculation engine.

**KEY FEATURES & CODE LOGIC:**
• Converts street addresses into latitude/longitude coordinates.
• Calculates driving distance and ETA between technician and customer site.
• Provides reverse geocoding for current GPS coordinates.

**DEVELOPER NOTE:** Used during booking step 3 and technician GPS navigation.

### FILE: src/utils/fileUpload.ts

**PRIMARY ROLE:** Image compression and base64 conversion utility.

**KEY FEATURES & CODE LOGIC:**
• Reads image file URIs from ImagePicker or Camera.
• Converts binary images to optimized Base64 data URLs for API transmission.

**DEVELOPER NOTE:** Prevents multipart/form-data boundary mismatch issues by sending standardized payloads.

## SECTION 5: CORE UI & SHARED COMPONENTS (SRC/COMPONENTS/) 10 Core Files

*Reusable presentation components, modals, headers, icon systems, and design system elements.*

### FILE: src/components/customer/RazorpayCheckoutModal.tsx

**PRIMARY ROLE:** Interactive Razorpay payment modal with live SDK injection.

**KEY FEATURES & CODE LOGIC:**
• Dynamically loads official Razorpay `checkout.js` on web.
• Initializes Razorpay Standard Checkout with custom brand colors and real Key ID.
• Handles UPI, Cards, Net Banking, and Wallets in live production gateway.
• Validates signature HMAC upon success and notifies parent invoice screen.
• Provides sandbox simulator fallback for offline testing.

**DEVELOPER NOTE:** Configured with live key `rzp_live_Svtu4hOoM1CKWE`. Guarded against duplicate order creation.

### FILE: src/components/common/AppIcon.tsx

**PRIMARY ROLE:** Unified cross-platform icon adapter wrapping Lucide & Expo Vector Icons.

**KEY FEATURES & CODE LOGIC:**
• Exposes 80+ standard icon glyphs (Wrench, ShieldCheck, MapPin, Tag, Banknote, User, etc.).
• Supports custom stroke colors, fill colors, and sizing.
• Falls back cleanly if an icon name is missing.

**DEVELOPER NOTE:** Used across all 80+ screens in the application for consistent UI iconography.

### FILE: src/components/common/ScreenContainer.tsx

**PRIMARY ROLE:** Universal page scaffold wrapper.

**KEY FEATURES & CODE LOGIC:**
• Applies automatic Safe Area insets (top, bottom, notch).
• Standardizes background surface colors across themes.
• Supports custom padding and scrolling wrappers.

**DEVELOPER NOTE:** Every top-level screen uses ScreenContainer for layout consistency.

### FILE: src/components/common/PhotoCaptureModal.tsx

**PRIMARY ROLE:** Diagnostic photo capture and attachment sheet.

**KEY FEATURES & CODE LOGIC:**
• Allows technician or customer to take live camera photos or choose from photo library.
• Categorizes photos with audit tags ('Pre-Repair Inspection', 'Hydraulic Cylinder', 'Post-Repair Audit').
• Compresses photos into data URLs.

**DEVELOPER NOTE:** Used during Active Job repair documentation and customer dispute submissions.

### FILE: src/components/common/LiveNavigationMap.tsx

**PRIMARY ROLE:** Live map visualizer with simulated vehicle telemetry.

**KEY FEATURES & CODE LOGIC:**
• Displays interactive map tile with technician van marker and customer location pin.
• Simulates realistic movement and recalculates live ETA based on velocity.
• Includes 'Open in Google Maps' button for external native navigation.

**DEVELOPER NOTE:** Renders seamlessly on web via SVG / Canvas and on native via map components.

### FILE: src/components/customer/DisputeTicketModal.tsx

**PRIMARY ROLE:** Customer dispute claim and escrow pause modal.

**KEY FEATURES & CODE LOGIC:**
• Presents dispute category selector (Workmanship, Punctuality, Overcharging, Damage).
• Accepts evidence description and attachment photos.
• Submits ticket to backend `/api/v1/customer/disputes` to halt funds release.

**DEVELOPER NOTE:** Pauses automatic technician payout until arbitration signoff.

### FILE: src/components/common/VoiceCallModal.tsx

**PRIMARY ROLE:** In-app simulated VoIP voice call interface.

**KEY FEATURES & CODE LOGIC:**
• Displays caller avatar, technician name, and call timer.
• Includes Mute, Speakerphone, and End Call actions.
• Plays audio feedback for connect and disconnect events.

**DEVELOPER NOTE:** Used on web where `tel:` links cannot place native cellular calls.

### FILE: src/components/dev/DevRoleSwitcher.tsx

**PRIMARY ROLE:** Developer role switcher overlay button.

**KEY FEATURES & CODE LOGIC:**
• Allows instant switching between Customer, Technician, and SuperAdmin views with 1 tap.
• Visible only in dev builds or when explicitly enabled.

**DEVELOPER NOTE:** Accelerates pair-programming and UI testing across roles.

### FILE: src/components/common/BrandLogo.tsx

**PRIMARY ROLE:** Standardized SVG brand logo emblem.

**KEY FEATURES & CODE LOGIC:**
• Renders premium glowing hexagonal ergonomic chair icon with gradient blue backdrop.
• Configurable size, border radius, and container styling.
• Integrated across all role headers: Customer, Technician, and SuperAdmin.

**DEVELOPER NOTE:** Replaces legacy image asset dependencies with high-resolution vector rendering.

### FILE: src/components/common/TechnicianHeaderAvatar.tsx

**PRIMARY ROLE:** Universal technician profile header avatar component.

**KEY FEATURES & CODE LOGIC:**
• Circular luxury avatar rendering with midnight navy (#0F172A) palette and bold white initials.
• Direct navigation jump to Technician Profile tab on tap.
• Optional verified specialist check badge overlay.

**DEVELOPER NOTE:** Standardized across all technician header bars for consistent branding.

## SECTION 6: CUSTOMER WORKFLOW SCREENS (SRC/SCREENS/CUSTOMER/)

**14 Core Files**

*20+ production screens covering the customer end-to-end journey from service booking to payment & review.*

### FILE: src/screens/customer/PaymentInvoiceChaircareMobileScreen.tsx

**PRIMARY ROLE:** Official digital tax invoice and Razorpay checkout trigger screen.

**KEY FEATURES & CODE LOGIC:**
• Single-Payment Guard: Prevents redundant charges if `is_paid === true`.
• Displays itemized parts, labor charges, BIFMA calibration rate, and tax breakdown.
• If already paid: Displays green 'PAID ONLINE • DO NOT PAY TECHNICIAN' banner with receipt ID.
• If unpaid: Launches live Razorpay interactive checkout modal.
• Shows 256-bit encryption and RBI compliance credentials.

**DEVELOPER NOTE:** Guaranteed strictly 1 payment per order. Disables pay buttons when already paid.

### FILE: src/screens/customer/ActiveServiceChaircareMobileScreen.tsx

**PRIMARY ROLE:** Live on-site repair execution tracking screen.

**KEY FEATURES & CODE LOGIC:**
• Live elapsed repair timer with SLA countdown.
• Shows technician profile, contact button, and mechanical checklist.
• Prominently reflects payment status: 'PAID ONLINE • DO NOT PAY TECH' vs 'PAY ON DELIVERY • DUE UPON COMPLETION'.
• Provides direct access to emergency support and chat.

**DEVELOPER NOTE:** Synchronizes in real time with the technician's active job progress.

### FILE: src/screens/customer/BookingStep1SelectServiceChaircareMobileScreen.tsx

**PRIMARY ROLE:** Step 1: Ergonomic service package selector.

**KEY FEATURES & CODE LOGIC:**
• Displays catalog: Precision Overhaul, Hydraulic Cylinder Replacement, Tilt Lock Fix, Wheel Caster Replacement.
• Shows base pricing, duration estimate, and BIFMA certification tags.

**DEVELOPER NOTE:** Initializes `draft.serviceTitle` and `draft.estimatedTotal` in BookingContext.

### FILE: src/screens/customer/BookingStep2DescribeProblemChaircareMobileScreen.tsx

**PRIMARY ROLE:** Step 2: Mechanical issue description & chair model details.

**KEY FEATURES & CODE LOGIC:**
• Accepts chair brand/model (Herman Miller Aeron, Steelcase Gesture, Featherlite, Generic).
• Input for mechanical symptoms (sinking gas lift, stuck tilt, broken caster).
• Diagnostic photo upload integration.

**DEVELOPER NOTE:** Stores problem description into the booking draft.

### FILE: src/screens/customer/BookingStep3ServiceLocationChaircareMobileScreen.tsx

**PRIMARY ROLE:** Step 3: Service site address and building entrance details.

**KEY FEATURES & CODE LOGIC:**
• Address input with floor/unit number, street name, and gate entry code.
• Interactive map picker modal integration with GPS pin drop.

**DEVELOPER NOTE:** Converts street address to geo-coordinates via geocodingService.

### FILE: src/screens/customer/BookingStep4ScheduleVisitChaircareMobileScreen.tsx

**PRIMARY ROLE:** Step 4: Date calendar and time slot selector.

**KEY FEATURES & CODE LOGIC:**
• Horizontal date strip (Today, Tomorrow, and upcoming 7 days).
• Morning, Afternoon, and Evening slot chips with live availability indicators.

**DEVELOPER NOTE:** Validates minimum 2-hour scheduling lead time.

### FILE: src/screens/customer/BookingStep5ConfirmRequestChaircareMobileScreen.tsx

**PRIMARY ROLE:** Step 5: Review summary, payment mode selection, and order placement.

**KEY FEATURES & CODE LOGIC:**
• Final booking audit card summarizing chair type, symptoms, address, and slot.
• Dynamic 10% Platform Fee: Automatically computed on base service pricing and itemized for full transparency.
• Choice between 'Prepaid Online via Razorpay' and 'Pay on Delivery (Cash/UPI)'.
• Official Support & Helpdesk: Direct support contact details (+91 92174 60742, enquiry@offheman.in).
• Dispatches request to backend via `customerApi.createServiceRequest()`.

**DEVELOPER NOTE:** Generates tracking code (e.g. CC-8492) and navigates to FindingATechnician.

### FILE: src/screens/customer/FindingATechnicianChaircareMobileScreen.tsx

**PRIMARY ROLE:** Radar sweep visualizer matching nearest available field technician.

**KEY FEATURES & CODE LOGIC:**
• Dynamic spinning radar sweep animation scanning 5 km radius.
• Live multi-step matching progress checklist.
• Polls backend for assignment acceptance every 3 seconds.

**DEVELOPER NOTE:** Transitions automatically to `TechnicianAssigned` as soon as a technician accepts.

### FILE: src/screens/customer/TechnicianAssignedChaircareMobileScreen.tsx

**PRIMARY ROLE:** Technician assignment confirmation and profile briefing.

**KEY FEATURES & CODE LOGIC:**
• Displays certified specialist photo, name, BIFMA rating, and vehicle details.
• Shows estimated arrival time (ETA) and direct phone/chat action buttons.
• Presents 4-digit Security PIN for customer to provide upon technician arrival.

**DEVELOPER NOTE:** The 4-digit PIN is required by the technician to close the job.

### FILE: src/screens/customer/LiveTechnicianTrackingChaircareMobileScreen.tsx

**PRIMARY ROLE:** Live GPS navigation and route tracking screen.

**KEY FEATURES & CODE LOGIC:**
• Embedded LiveNavigationMap showing live specialist van moving toward customer location.
• Distance remaining (km) and live arrival countdown (mins).
• Emergency dispatch helpline button.

**DEVELOPER NOTE:** Simulates smooth waypoint interpolation between tech location and customer site.

### FILE: src/screens/customer/ServiceCompletedChaircareMobileScreen.tsx

**PRIMARY ROLE:** Post-service sign-off, invoice review, and warranty activation.

**KEY FEATURES & CODE LOGIC:**
• Celebratory completion banner with 90-day mechanical warranty badge.
• Shows before/after inspection photos signed off by technician.
• Button to download PDF tax invoice or view payment receipt.

**DEVELOPER NOTE:** Directs customer to RateYourExperience screen.

### FILE: src/screens/customer/MyRequestsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Central customer booking history and active requests list.

**KEY FEATURES & CODE LOGIC:**
• Segmented tabs: 'Active', 'Upcoming', 'Completed'.
• Renders request cards with live status badges.
• Accurately shows 'Paid Online' (green) or 'Pay on Delivery' (amber) for every order.

**DEVELOPER NOTE:** Pulled from `GET /api/v1/customer/requests` with pull-to-refresh.

### FILE: src/screens/customer/RequestDetailsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Comprehensive deep-dive view of any single service request.

**KEY FEATURES & CODE LOGIC:**
• Full chronology timeline: Request Submitted -> Tech Assigned -> On-Site -> Completed.
• Itemized pricing card reflecting 'Total Paid (Online)' vs 'Pay on Delivery'.
• Button to raise dispute ticket or download official service PDF.

**DEVELOPER NOTE:** Connected to `customerApi.getRequestDetails(requestId)`.

### FILE: src/screens/customer/ChatWithTechnicianChaircareMobileScreen.tsx

**PRIMARY ROLE:** Direct encrypted messaging screen between customer and assigned technician.

**KEY FEATURES & CODE LOGIC:**
• Live chat bubble stream with timestamps and delivery checkmarks.
• Image attachment sharing for mechanical damage photos.
• Integrated phone call shortcut.

**DEVELOPER NOTE:** Backed by `chatApi` and database table `chat_messages`.

## SECTION 7: TECHNICIAN PLATFORM SCREENS (SRC/SCREENS/TECHNICIAN/)

**10 Core Files**

*Mobile screens for field technicians to receive jobs, navigate, calibrate chairs, verify PINs, and track payouts.*

### FILE: src/screens/technician/ActiveJobChaircareMobileScreen.tsx

**PRIMARY ROLE:** Primary on-site job execution checklist and live telemetry screen.

**KEY FEATURES & CODE LOGIC:**
• High-Visibility Collection Directive: Prominently warns technician 'PAID ONLINE • DO NOT COLLECT MONEY' vs 'PAY ON DELIVERY • COLLECT ■{amount}'.
• Live elapsed repair stopwatch with equalizer animation.
• Photo audit logger capturing pre-repair and post-repair mechanical photos.
• Customer 4-digit Security PIN verification modal before job sign-off.

**DEVELOPER NOTE:** Crucial screen for field operations. Passes `isPaid` state cleanly to `JobCompleted`.

### FILE: src/screens/technician/JobCompletedChaircareMobileScreen.tsx

**PRIMARY ROLE:** Post-job signoff, customer payment QR presentation, and earnings settlement confirmation.

**KEY FEATURES & CODE LOGIC:**
• Post-OTP Payment QR Generator: 'Show Customer Payment QR' button presents dynamic scannable UPI QR code.
• Embedded UPI Deep Links: Automatically encodes technician's UPI VPA and exact grand total into standard UPI URI.
• Instant Payment Receipt Confirmation: Technician confirms on-site UPI payment, automatically marking order as paid (`is_paid = true`).
• Direct Payment Logging: Adds payment transaction directly into technician's local and cloud earnings history.
• Warranty Activation & Signoff: 90-day mechanical warranty badge and feedback prompt.

**DEVELOPER NOTE:** Seamlessly transitions completed jobs to paid status and logs direct UPI collections.

### FILE: src/screens/technician/JobRequestsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Nearby unassigned job requests dispatch queue.

**KEY FEATURES & CODE LOGIC:**
• Displays incoming jobs within technician's service radius with distance and payout.
• Shows customer address, chair model, and payment mode (Prepaid vs POD).
• Swipe-to-accept or tap-to-review actions.

**DEVELOPER NOTE:** Polls `GET /api/v1/technician/available-jobs`.

### FILE: src/screens/technician/JobRequestDetailsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Comprehensive inspection of a job request before accepting.

**KEY FEATURES & CODE LOGIC:**
• Full problem description, photos, customer location map, and payout breakdown.
• High-visibility banner clarifying whether order is Paid Online or Pay on Delivery.
• Action button to accept request and claim dispatch.

**DEVELOPER NOTE:** Accepting calls `POST /api/v1/technician/accept-job` and assigns job to current technician.

### FILE: src/screens/technician/NavigateToCustomerChaircareMobileScreen.tsx

**PRIMARY ROLE:** Turn-by-turn navigation route guidance toward customer address.

**KEY FEATURES & CODE LOGIC:**
• Map displaying technician van moving toward destination.
• Gate code reminder and customer phone shortcut.
• Button to transition to 'Arrived at Customer'.

**DEVELOPER NOTE:** Updates order status to `in_route` on backend.

### FILE: src/screens/technician/ArrivedAtCustomerChaircareMobileScreen.tsx

**PRIMARY ROLE:** On-site arrival confirmation and customer greeting checklist.

**KEY FEATURES & CODE LOGIC:**
• Notifies customer that technician has arrived at the door.
• Presents 5-point BIFMA on-site arrival safety checklist.
• Button to start diagnostic inspection and begin repair timer.

**DEVELOPER NOTE:** Transitions status to `arrived` then `in_progress`.

### FILE: src/screens/technician/AdditionalWorkApprovalChaircareMobileScreen.tsx

**PRIMARY ROLE:** On-site scope expansion and extra parts approval request.

**KEY FEATURES & CODE LOGIC:**
• Allows technician to request extra parts (e.g. Heavy Duty Gas Cylinder +■450).
• Sends push notification to customer for real-time digital approval before installation.

**DEVELOPER NOTE:** Protects customers from unexpected charges without prior authorization.

### FILE: src/screens/technician/EarningsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Technician earnings dashboard, direct UPI QR collections, and payout ledger.

**KEY FEATURES & CODE LOGIC:**
• Admin Reference Notice: Explicitly notes dummy ledger records are for admin reference only.
• Direct UPI Collection Hub: Displays technician's configured UPI ID, linked phone, and standby scannable QR modal.
• Instant Dynamic UPI Edit Modal: Technicians can modify and test their UPI VPA and phone number dynamically.
• On-Site QR Collection Ledger: Records all real-time direct payments collected from customers via QR.
• Historical earnings analytics: Daily, weekly, and monthly totals with payout status.

**DEVELOPER NOTE:** Reads direct QR collections from AsyncStorage and dynamic UPI credentials from TechnicianContext.

### FILE: src/screens/technician/TechnicianDashboardChaircareMobileScreen.tsx

**PRIMARY ROLE:** Main home command center for logged-in technician.

**KEY FEATURES & CODE LOGIC:**
• Availability status toggle (Online / Offline for dispatches).
• Active job shortcut card with current status and customer name.
• Performance metrics: Completion Rate, Average Rating (4.9★), Total Earned.

**DEVELOPER NOTE:** Home tab of TechnicianNavigator.

### FILE: src/screens/technician/TechnicianProfileChaircareMobileScreen.tsx

**PRIMARY ROLE:** Field technician profile, avatar customization, and direct UPI payout settings.

**KEY FEATURES & CODE LOGIC:**
• Certified Specialist Profile: Verified Fleet Specialist badge, completed job count, and average rating.
• Luxury Circular Avatar: Executive midnight navy (#0F172A) profile avatar with white initials and camera edit badge.
• Bank & Payout Details Modal: Dynamic UPI ID and UPI phone number management with live scannable QR preview.
• Operational Corridor & Specializations: Configurable service radius (15 km) and chair mechanical specialties.
• Fleet Support Hotline: Direct shortcuts to official helpline (+91 92174 60742) and email (enquiry@offheman.in).

**DEVELOPER NOTE:** Synchronizes profile changes and UPI payout credentials directly with backend API.

## SECTION 8: OPERATIONS & SUPERADMIN SCREENS (SRC/SCREENS/ADMIN/) — 8 Core Files

*40+ administrative management screens for fleet dispatchers, dispute arbitrators, and business administrators.*

### FILE: src/screens/admin/LiveOperationsOverviewChaircareMobileScreen.tsx

**PRIMARY ROLE:** High-altitude live command overview of platform health.

**KEY FEATURES & CODE LOGIC:**
• Live telemetry tiles: Active Jobs, Open Requests, Online Technicians, Daily Gross Revenue.
• Real-time dispatch alert feed and urgent escalations list.
• Quick filters for delayed jobs or disputed orders.

**DEVELOPER NOTE:** Connects to `GET /api/v1/admin/dashboard`.

### FILE: src/screens/admin/ServiceRequestsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Master dispatch ledger of all platform service tickets.

**KEY FEATURES & CODE LOGIC:**
• Filterable list by status: Unassigned, In-Route, Active, Completed, Cancelled.
• Shows payment state (Paid Online vs Pay on Delivery), customer details, and assigned tech.
• Manual override button to reassign any ticket to a different technician.

**DEVELOPER NOTE:** Full CRUD management for dispatchers.

### FILE: src/screens/admin/TechniciansChaircareMobileScreen.tsx

**PRIMARY ROLE:** Fleet registry of all registered and certified technicians.

**KEY FEATURES & CODE LOGIC:**
• Shows KYC status (Verified, Pending KYC, Suspended).
• Displays ratings, completed job counts, and current availability.
• Buttons to verify Aadhaar/PAN documents or suspend technician account.

**DEVELOPER NOTE:** Updates `users` and `technician_profiles` tables.

### FILE: src/screens/admin/PaymentsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Financial transaction ledger and gateway reconciliation.

**KEY FEATURES & CODE LOGIC:**
• Records all Razorpay transactions with payment ID, order ID, amount, and timestamp.
• Filters by successful, pending, and refunded transactions.
• Action to initiate immediate customer refund with payment gateway sync.

**DEVELOPER NOTE:** Connected to `backend/src/controllers/adminController.ts` payment endpoints.

### FILE: src/screens/admin/DisputesChaircareSuperAdminScreen.tsx

**PRIMARY ROLE:** Dispute claims adjudication and escrow arbitration board.

**KEY FEATURES & CODE LOGIC:**
• Lists open customer dispute tickets with reason, photos, and amount held.
• Allows admin to review chat logs, inspect repair photos, and contact both parties.
• Decisions: Release payout to technician, issue full refund to customer, or partial settlement.

**DEVELOPER NOTE:** Unpauses or refunds payments held in platform escrow.

### FILE: src/screens/admin/ReviewsRatingsChaircareMobileScreen.tsx

**PRIMARY ROLE:** Review moderation and customer feedback management.

**KEY FEATURES & CODE LOGIC:**
• Lists all 1-star to 5-star customer reviews across all technicians.
• Allows superadmin to hide abusive reviews or flag recurring mechanical failures.

**DEVELOPER NOTE:** Protects platform reputation and technician feedback integrity.

### FILE: src/screens/admin/AuditExportChaircareMobileScreen.tsx

**PRIMARY ROLE:** System audit logging and financial data export engine.

**KEY FEATURES & CODE LOGIC:**
• Export service requests, technician earnings, and tax invoices as CSV/PDF.
• Date range filter and compliance verification stamps.

**DEVELOPER NOTE:** Generates data exports for accounting and tax compliance.

### FILE: src/screens/admin/PlatformConfigurationChaircareMobileScreen.tsx

**PRIMARY ROLE:** Dynamic platform settings, service pricing tiers, and commission rate configuration.

**KEY FEATURES & CODE LOGIC:**
• Configure base diagnostic fee and dynamic platform fee percentage (10% default).
• Service Pricing Configuration: Active standard tiers with deprecated same-day tier cleanly removed.
• Set dispatch radius (5 km / 10 km) and technician matching timeout thresholds.

**DEVELOPER NOTE:** Updates `platform_configurations` table dynamically without app redeployment.

## SECTION 9: BACKEND CORE ARCHITECTURE (BACKEND/SRC/) 14 Core Files

*Node.js TypeScript Express REST API server, PostgreSQL database pool, Razorpay integration, and migrations.*

### FILE: backend/src/server.ts

**PRIMARY ROLE:** HTTP server entry point and port listener.

**KEY FEATURES & CODE LOGIC:**
• Imports Express application from `app.ts`.
• Checks database connection pool health on startup.
• Listens on environment `PORT` (default 5000 / 10000 on Render).
• Implements graceful shutdown on SIGTERM/SIGINT signals.

**DEVELOPER NOTE:** Bootstraps backend service in production.

### FILE: backend/src/app.ts

**PRIMARY ROLE:** Express application setup, middleware pipeline, and route registration.

**KEY FEATURES & CODE LOGIC:**
• Configures CORS, Helmet security headers, body parsing (10MB JSON for photos).
• Mounts API route branches: `/auth`, `/customer`, `/technician`, `/admin`, `/payments`, `/chat`, `/services`, `/notifications`.
• Integrates global rate limiter and centralized error handling middleware.

**DEVELOPER NOTE:** Add any new API route modules in this file.

### FILE: backend/src/config/database.ts

**PRIMARY ROLE:** PostgreSQL connection pool management via `pg.Pool`.

**KEY FEATURES & CODE LOGIC:**
• Configures connection pooling with SSL rejectUnauthorized: false for cloud DBs (Supabase / Render).
• Exposes helper `query(text, params)` with automatic client checkout and release.
• Logs query execution latency in development.

**DEVELOPER NOTE:** All controllers interact with PostgreSQL via this query abstraction.

### FILE: backend/src/config/env.ts

**PRIMARY ROLE:** Environment variable validation and parsing.

**KEY FEATURES & CODE LOGIC:**
• Validates `DATABASE_URL`, `JWT_SECRET`, `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET`.
• Provides typed configuration object throughout backend.

**DEVELOPER NOTE:** Fails fast at server startup if critical secrets are missing.

### FILE: backend/src/controllers/paymentController.ts

**PRIMARY ROLE:** Razorpay order creation, signature verification, and single-payment enforcement.

**KEY FEATURES & CODE LOGIC:**
• Enforces 1-Order-1-Payment: Rejects duplicate order creation if request is already paid.
• Creates official Razorpay Order via `razorpayService.createOrder` with currency INR.
• Verifies cryptographic HMAC-SHA256 signature on `/api/v1/payments/verify`.
• Idempotently updates `service_requests.is_paid = true` and logs to `payments` table.
• Logs payment audit trail to `admin_activity_logs` and sends customer notification.

**DEVELOPER NOTE:** Guarantees zero double-charging and keeps payments strictly 1-to-1 with orders.

### FILE: backend/src/controllers/customerController.ts

**PRIMARY ROLE:** Customer service request lifecycle controller.

**KEY FEATURES & CODE LOGIC:**
• Handles `POST /requests`: creates order, generates tracking code, assigns initial status.
• Handles `GET /requests`: returns customer's booking history with technician details and is_paid flag.
• Handles `GET /requests/:id`: returns deep ticket chronology with assigned specialist details.
• Handles cancellation, customer rating submission, and dispute ticket creation.

**DEVELOPER NOTE:** Handles non-UUID service codes gracefully with fallback defaults.

### FILE: backend/src/controllers/technicianController.ts

**PRIMARY ROLE:** Technician job queue, navigation, PIN completion, and dynamic UPI payout controller.

**KEY FEATURES & CODE LOGIC:**
• Returns unassigned jobs within radius (`getAvailableJobs`).
• Accepts job assignment atomically (`acceptJob`).
• Updates technician live GPS location (`updateLocation`).
• Verifies customer's 4-digit Security PIN (`verifyPin`) to complete repair.
• Dynamic UPI Payout Credentials: Manages technician `upiId` and `upiPhone` via `getProfile` and `updateTechnicianProfile`.
• Direct UPI Payment Sync: Marks order as `is_paid = true` upon direct customer UPI payment confirmation.
• Handles photo uploads, additional work requests, and earnings calculations.

**DEVELOPER NOTE:** Atomic database transactions prevent collisions; UPI credentials stored in `technician_profiles`.

### FILE: backend/src/controllers/adminController.ts

**PRIMARY ROLE:** SuperAdmin analytics, user management, and financial control.

**KEY FEATURES & CODE LOGIC:**
• Returns platform-wide KPIs: active orders, gross GMV, fleet availability.
• Manages technician approvals, KYC verification, and account suspensions.
• Processes customer refunds with Razorpay API sync and dispute resolution.
• Exports CSV audit datasets for operations and accounting.

**DEVELOPER NOTE:** Protected by `verifyToken` and `requireAdmin` middleware.

### FILE: backend/src/controllers/authController.ts

**PRIMARY ROLE:** User registration, authentication, and JWT generation.

**KEY FEATURES & CODE LOGIC:**
• Hashes passwords using Bcrypt (salt rounds 10).
• Issues signed JWT tokens containing `userId`, `email`, and `role`.
• Provides role-based login for customers, technicians, and administrators.
• Supports password change and profile updates.

**DEVELOPER NOTE:** Token expires in 7 days. Verified on every incoming request.

### FILE: backend/src/controllers/chatController.ts

**PRIMARY ROLE:** In-app messaging controller between customers and technicians.

**KEY FEATURES & CODE LOGIC:**
• Stores text messages and image attachments in `chat_messages` table.
• Retrieves chronologically ordered message streams by service request ID.
• Marks messages as read when recipient opens conversation.

**DEVELOPER NOTE:** Includes unread message counter for tab badge updates.

### FILE: backend/src/services/razorpayService.ts

**PRIMARY ROLE:** Direct interface with Razorpay Orders API and crypto verification.

**KEY FEATURES & CODE LOGIC:**
• Wraps official `razorpay` Node SDK initialized with live Key ID and Secret.
• Creates orders with rupee-to-paise conversion (`amount * 100`).
• Calculates HMAC-SHA256 signature (`crypto.createHmac`) to verify webhook/callback integrity.
• Includes sandbox simulator fallback when running without production credentials.

**DEVELOPER NOTE:** Configured via environment: `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET`.

### FILE: backend/src/middleware/auth.ts

**PRIMARY ROLE:** JWT authorization and role-based access control (RBAC).

**KEY FEATURES & CODE LOGIC:**
• `verifyToken`: Validates Bearer token in HTTP header and attaches `req.user`.
• `requireRole(['customer', 'technician', 'admin'])`: Restricts endpoints to authorized roles.
• `requireAdmin`: Dedicated guard for SuperAdmin management endpoints.

**DEVELOPER NOTE:** Returns 401 Unauthorized or 403 Forbidden on invalid tokens.

### FILE: backend/src/validators/schemas.ts

**PRIMARY ROLE:** Zod request body and parameter validation schemas.

**KEY FEATURES & CODE LOGIC:**
• `createRequestSchema`: Validates service creation inputs, dates, and addresses.
• `loginSchema` & `registerSchema`: Validates email formats and password lengths.
• `verifyPaymentSchema`: Validates Razorpay payment ID, order ID, and signature strings.

**DEVELOPER NOTE:** Intercepts malformed requests before they hit controller database queries.

**FILE: backend/src/migrations/001_initial_schema.sql to 018_technician_upi_payout_details.sql**

**PRIMARY ROLE:** PostgreSQL database schema migrations.

**KEY FEATURES & CODE LOGIC:**
• 001: Core tables (`users`, `services`, `service_requests`, `payments`, `reviews`).
• 002-005: Notification system, chat message logs, 4-digit Security PIN verification.
• 006-009: Technician KYC verification, real-time GPS coordinates.
• 010-016: Repair timer, photo audit logs, support chat, rate cards, dispatch rules.
• 017: Platform pricing configuration, dynamic platform commission fee, and service tiers.
• 018: Technician direct UPI payout credentials (`upi_id`, `upi_phone`) on `technician_profiles`.

**DEVELOPER NOTE:** Executed sequentially by `migrations/runMigrations.ts` or standalone script against cloud PostgreSQL.